ArchiLudo

A Ludo ("Mens Erger Je Niet") game for the Acorn Archimedes, targeting genuine 26-bit RISC OS 3.10. By Xander Mol (idreamtin8bits.com).

Contents

Playing ArchiLudo

Four players, four pawns each, a shared ring and a short home run per player – the classic "Mens Erger Je Niet"/Ludo cross board.

Starting a game: on first launch, dismissing the startup splash screen (**OK**, or click anywhere on it) opens the **New Game** dialogue directly. Later on, click the iconbar icon to open it again. Set each of the 4 players' name and whether they're Human or AI-controlled; an AI-controlled player also gets a **Low**/**Medium**/ **High** difficulty picker (click to cycle) – see AI.md for what each tier actually does. Optionally click **Rules...** to pick a rule variant or adjust individual house rules (see RULES.md for the full manual), then **Start**.

Playing a turn: click **Throw** to roll. The status line tells you what to do next – click one of your own pawns on the board if more than one can legally move (if only one can, it moves automatically), or click **Throw** again on a six. Landing exactly on another pawn sends it home; reaching the exact end of your home column finishes that pawn. For an AI player's turn, click **Continue** to step through its moves. When someone gets all four pawns home, a win screen appears – **Continue** lets the remaining players keep going to decide runner-up order.

Saving and loading: **Save Game**/**Load Game** from the iconbar menu open 5 fixed, renamable slots at any time during play.

Music and sound effects: open the **Music** submenu (iconbar or window menu) to turn music **On**/off, turn **SFX** on/off independently, or pick a **Track**. See QTM.md for the full music/SFX manual (what each track and effect is, and Track 3's one limitation).

Rules and variants: ArchiLudo ships 3 rule presets (Mens Erger Je Niet, Ludo, Pachisi-style) and 8 individually-adjustable house-rule toggles. See RULES.md for the complete rules manual.

Installation

Requires a real or emulated Acorn Archimedes running RISC OS 3.10 or later (ARM2 or ARM3). Download a release (`ArchiLudo-vX.Y.Z-*.zip` or `.adf`) and:

- **Zip**: extract with SparkFS (or any RISC-OS-aware zip tool) – filetypes are preserved. Double-click the extracted `!ArchiLudo` application directory to run.
- **Disc image** (`.adf`, ADFS "D" format, 800KB): write it to a real 3.5" floppy, or mount it directly in an emulator that supports raw ADFS images. It contains just the release zip, correctly filetyped – extract it the same way as above once it's accessible from RISC OS.
- **Real classic-Econet hardware** (a PiEconetBridge or similar old-style fileserver): rename the downloaded file to 10 characters or fewer with no dot before transferring it (e.g. `ArchiZip` for the zip, `ArchiADF` for the disc image) – such fileservers silently truncate longer names in a way that makes the file unreadable, not just renamed. Arculator's hostfs and a plain download/extract on Windows/Mac/Linux have no such limit.

No installer is needed beyond copying the `!ArchiLudo` directory wherever you want it – RISC OS applications are self-contained directories, not registry entries or system-wide installs.

Building from source

Prerequisites

Tool	Purpose	Install
ArchieSDK	ARM2-targeting cross-compiler (GCC 8.5.0), bundled OSLib	`git clone https://gitlab.com/_targz/archiesdk.git ~/riscos-dev/archiesdk && cd ~/riscos-dev/archiesdk && ./build.sh`
Arculator	Acorn Archimedes emulator, for testing	already installed at `D:\Retro\Acorn\Arculator_V2.2_Windows`
pandoc	optional: README.md -> README.pdf	`sudo apt install pandoc texlive-xetex`
Python 3 + Pillow	`tools/riscos_sprite.py`, the PNG->Sprite converter	`pip install Pillow`
sshpass	optional: `make deploy-pibridge` (real-hardware deploy)	`sudo apt install sshpass`

.env setup

Copy `.env.example` to `.env` and set:

```
ARCHIESDK = /home/xahmol/riscos-dev/archiesdk
ARCULATOR_HOSTFS = /mnt/d/Retro/Acorn/Arculator_V2.2_Windows/hostfs
```

`PIBRIDGE_USER`/`PIBRIDGE_HOST`/`PIBRIDGE_PASS`/`PIBRIDGE_PATH` are also needed for `make deploy-pibridge` (a real PiEconetBridge target) – see `.env.example` for the full set. `.env` is gitignored – never commit it.

Make targets

Target	Effect
`make` / `make all`	build the `build/!ArchiLudo` application directory
`make test`	build and run the game-logic unit tests with the host compiler (no ArchieSDK/Arculator needed)
`make clean`	remove `build/`
`make deploy`	copy `build/!ArchiLudo` to the Arculator hostfs folder
`make deploy-pibridge`	deploy to a real PiEconetBridge over SSH (password auth)
`make zip`	versioned release archive (`build/ArchiLudo-vX.Y.Z-<timestamp>.zip`), RISC OS filetypes preserved
`make disk`	ADFS "D" format (800KB) disc image containing the release zip, correctly filetyped
`make assets`	regenerate the pawn sprites and app icon from their Python generators
`make export-sprites` / `make import-sprites`	round-trip shipped sprites to hand-editable PNGs and back
`make docs`	regenerate `README.pdf` via pandoc
`make asm`	emit generated ARM assembly for the current sources

See the "Installation" section above for the 10-character filename limit on real classic-Econet hardware – `make zip`/`make disk` deliberately keep the full version+timestamp in their own output filenames so multiple local builds can be told apart.

Testing: boot `configs/ArchiLudo-ARM3-4MB.cfg` (matches real ARM3/4MB hardware) or `configs/ArchiLudo-ARM2-1MB.cfg` (stock ARM2/1MB compatibility check) in Arculator with the RISC OS 3.10 ROM, then double-click

`!ArchiLudo` from HostFS via the Filer (see BUILDCHAIN.md's "Application directory" section for its structure).

Changelog

ArchiLudo hasn't cut discrete numbered releases yet – this is a milestone-level history of what's been built, newest first.

- **AI difficulty tiers**: Low/Medium/High per AI-controlled player, picked in the New Game dialogue – Low takes only the objectively decisive moves (win/finish/capture/progress), Medium is the original full heuristic, High adds a genuine one-ply lookahead. See AI.md.
- **Launch flow**: the New Game dialogue now opens automatically right after dismissing the startup splash screen.
- **Distribution pipeline**: a filetype-preserving release zip (`make zip`), a from-scratch ADFS disc-image builder (`make disk`), and a real-hardware deploy target over Econet (`make deploy-pibridge`) – all verified against real hardware, not just Arculator, including finding and fixing a 10-character filename limit on classic-Econet fileservers and a line-ending mismatch in the bundled plain-text README.
- **QTM audio**: background music (3 selectable tracks) and 6 one-shot sound effects, embedded as MOD instrument samples and triggered via `QTM_PlaySample`, independently toggleable, with loudness-normalized SFX. Confirmed working on real hardware.
- **Save/load rework**: from a PRM-correct but non-functional (on Arculator's HostFS) drag-and-drop design to 5 fixed, renamable save slots.
- **Multi-rule-set / house-rule variant system**: 3 curated presets (Mens Erger Je Niet, Ludo, Pachisi-style) plus 8 independently toggleable house rules, a Rule Options dialogue, and AI support for all of them – see RULES.md.
- **Win/continue flow**: a win no longer ends the game outright – players can continue for runner-up order or start a new game.
- **Sprite art pivot**: from `os_plot` primitives to real `Wimp_PlotIcon`-based pawn sprites and a proper application icon, plus a hand pixel-editing round-trip workflow for touching up art in an external editor.
- **AI opponents and player setup**: a "New Game" dialogue (names, Human/AI toggle per player) and a first AI opponent adapted from the original GeoLudo edition's own algorithm.
- **Animation and redraw overhaul**: flicker-free small-region animation (dice roll, pawn slide, pulsing highlights) via careful `Wimp_UpdateWindow` usage.
- **Application directory packaging**: a real `!ArchiLudo` app directory with a custom icon, replacing an earlier flat hostfs layout.
- **Phase 1: playable core loop**: the WIMP game window, the real Mens Erger Je Niet board (ported from the original GEOS edition's own coordinate tables), click-to-move, dice, capture/win detection.

- **Build environment**: the ArchieSDK toolchain, Arculator test profiles, and this project's full documentation set established.

Development history

ArchiLudo isn't this game's first outing – it's the latest in a line of ports going back over three decades, all by the same author. Full credits and technical detail for each generation are in `CREDITS.md`'s "Porting source" section; this is the narrative version.

- **1992**: the original, written in Commodore BASIC 7.0 for the Commodore 128 in 80-column mode – "Mens Erger Je Niet" (Dutch for "Don't Get Angry, Man", the house-rule variant of Ludo this whole family of ports implements), by Xander Mol under the "XAMA Software" name. The listing itself still carries its original 1992 copyright banner. Source and a working D64 disk image are preserved at `C128/BASIC Original` in the `xahmol/ludo` repository.
- **2020**: rewritten for the Oric Atmos, in BASIC.
- **2021**: rewritten a further three times – the Oric Atmos version again, now in C via CC65; a new C port for the TI-99/4a via TMS9900-GCC; and a new C port for the Commodore 128 via CC65.
- **2023**: **GeoLudo**, a GEOS edition sharing one executable across GEOS on the Commodore 64, Commodore 128, and Commodore Plus/4 – `xahmol/ludo/GEOS`. This is ArchiLudo's own direct porting source: its board geometry and ring/home-column layout, its pawn and die-face artwork, and its first AI opponent all trace back to GeoLudo specifically (see `docs/ARCHITECTURE.md`'s GeoLudo→ArchiLudo mapping table), even though ArchiLudo's own rules engine (`src/game_logic.c`) is a clean reimplementation rather than a line-for-line port (see `docs/GAME_LOGIC.md` for why).
- **ArchiLudo**: this project – a native WIMP port for the Acorn Archimedes, targeting genuine 26-bit RISC OS 3.10. The house rules the 1992 original encoded by hand are now the configurable `LUDO_VARIANT_MEJN` preset among three curated variants (see `RULES.md`), and its own multi-decade thread continues in this README's Changelog above.

Documentation

See `docs/` for the full project documentation: `ARCHITECTURE.md` (layering, directory structure, porting notes), `RULES.md` (the full player-facing rules manual), `GAME_LOGIC.md` (rules engine API), `BOARD_LAYOUT.md` (board geometry), `AI.md` (AI design), `QTM.md` (music/SFX manual and audio API), `GRAPHICS_TOOLING.md` (sprite converter and RISC OS sprite format), `BUILDCHAIN.md` (toolchain, Makefile, distribution), `OSLIB.md` and `LIBARCHIE.md` (the libraries this project builds against); plus `riscos_wimp_reference.md` for the WIMP/SWI/ message API reference, and `CREDITS.md` for everything this project is built on or derived from.

Credits and licence

See `CREDITS.md` for everything ArchiLudo is built on, ported from, or otherwise sourced from. Licensed under the GNU GPLv3.